

Mystery Dataset Investigation

NorthLedger Insights - automated analysis

Verdict: `mystery\_a` is the NASA GISS Surface Temperature Analysis (GISTEMP v4) - global surface temperature anomalies by year, 1880-2025, in degrees Celsius, relative to the 1951-1980 baseline. Confidence: very high.

Headline numbers: Warming from the 1880-1909 mean (-0.26 degC) to the 2016-2025 mean (+1.02 degC) = +1.28 degC. 2024 was the warmest year on record at +1.29 degC; 2023's jump of +0.28 degC was the largest single-year spike in the record. A 10-year forecast (fit on 1970-2025) projects +1.06 degC (2026) rising to +1.24 degC (2035), 80% band roughly +/-0.25-0.4 degC - but it does not beat the naive baseline in backtest, so treat it as directional only.

Mystery Dataset Investigation

1. What I saw in profiling

The table `mystery\_a` has 147 rows and 14 columns: `year`, `jan` ... `dec`, `annual`.

- `year` runs 1880 -> 2026, one row per year (distinct count = 147). The 2026 row is blank (future placeholder).
- Every monthly column holds small decimal numbers between about -0.8 and +1.3.
- No categorical text, no IDs, no null-like strings - a clean numeric time matrix.
- Pattern: values are negative in the 1800s, hover near zero mid-20th century, and turn strongly positive after ~1980, reaching ~+1.3 by 2024.

That structure - a year column plus 12 monthly and 1 annual column, values centered on zero that trend upward - is the signature of a monthly temperature anomaly series, not raw temperatures (raw monthly temps would be ~0-30 degC and not centered on zero).

2. Hypothesis

Hypothesis: this is a global surface temperature anomaly dataset - the annual values represent how much warmer/cooler each year was than a mid-20th-century reference, in degrees Celsius.

Supporting clue: the 2024 value is +1.29 and 2023 is +1.17. Those exact figures are widely reported for the last two record years.

3. Web confirmation

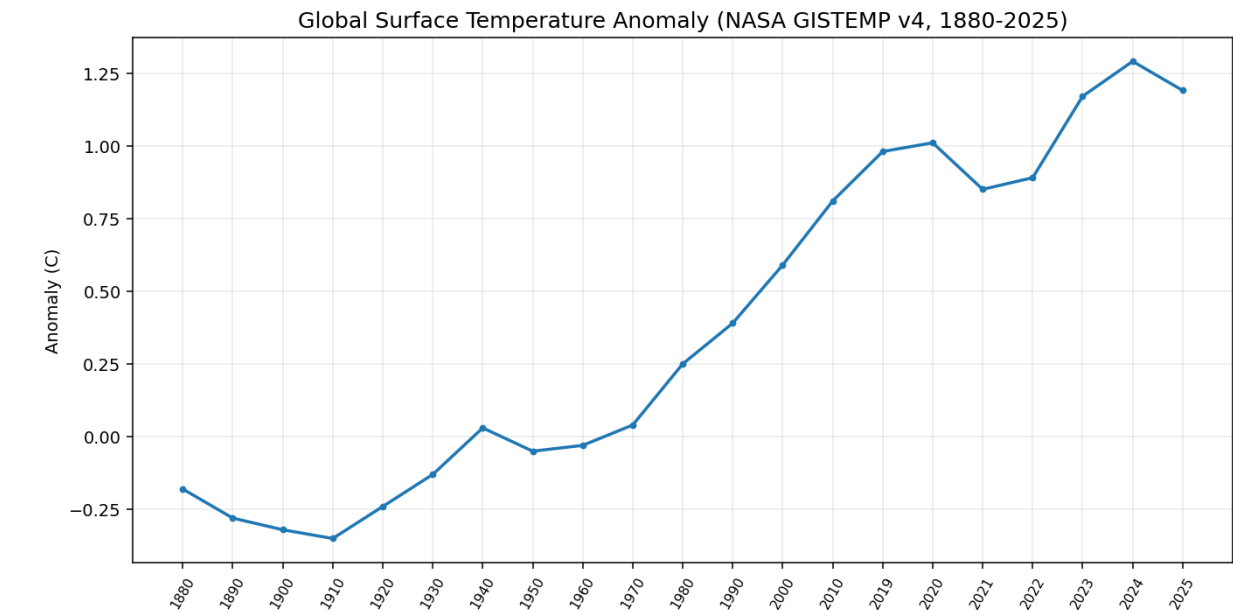
Searching for the column structure and values confirmed the identity:

- The NASA GISS Surface Temperature Analysis (GISTEMP v4) provides "monthly surface temperature anomaly data ... deviations from the 1951-1980" baseline [GISS Surface Temperature Analysis (GISTEMP v4) - <https://data.giss.nasa.gov/gistemp/>].
- A data mirror describes "Monthly global mean temperature anomalies in degrees Celsius from ... GISTEMP (1880-present)" [Global Temperature Time Series - <https://datahub.io/core/global-temp/>].
- NASA's 2024 release states 2024 was the warmest year on record, ~1.28-1.29 degC above the 1951-1980 baseline [Temperatures Rising: NASA Confirms 2024 Warmest Year on Record - <https://www.nasa.gov/news-release/temperatures-rising-nasa-confirms-2024-warmest-year-on-record/>].

Conclusion - Dataset identity:

Attribute	Value	Confidence
What it is	NASA GISTEMP v4 global surface temperature anomalies	Very high
Source	NASA Goddard Institute for Space Studies	Very high
Units	Degrees Celsius (degC)	Very high
Baseline	1951-1980 mean	High
Coverage	Annual, 1880-2025 (2026 blank)	Very high

4. Trend analysis



Decadal means make the trajectory unmistakable (from the `annual` column):

Decade	Mean anomaly (degC)
1880s	-0.22
1900s	-0.32
1910s	-0.34
1940s	+0.04
1960s	-0.03
1980s	+0.25
2000s	+0.59
2010s	+0.81
2020s	+1.07

Early vs recent (exact): mean of 1880-1909 = -0.261 degC; mean of 2016-2025 = +1.016 degC -> net change of +1.277 degC over ~1.1 centuries.

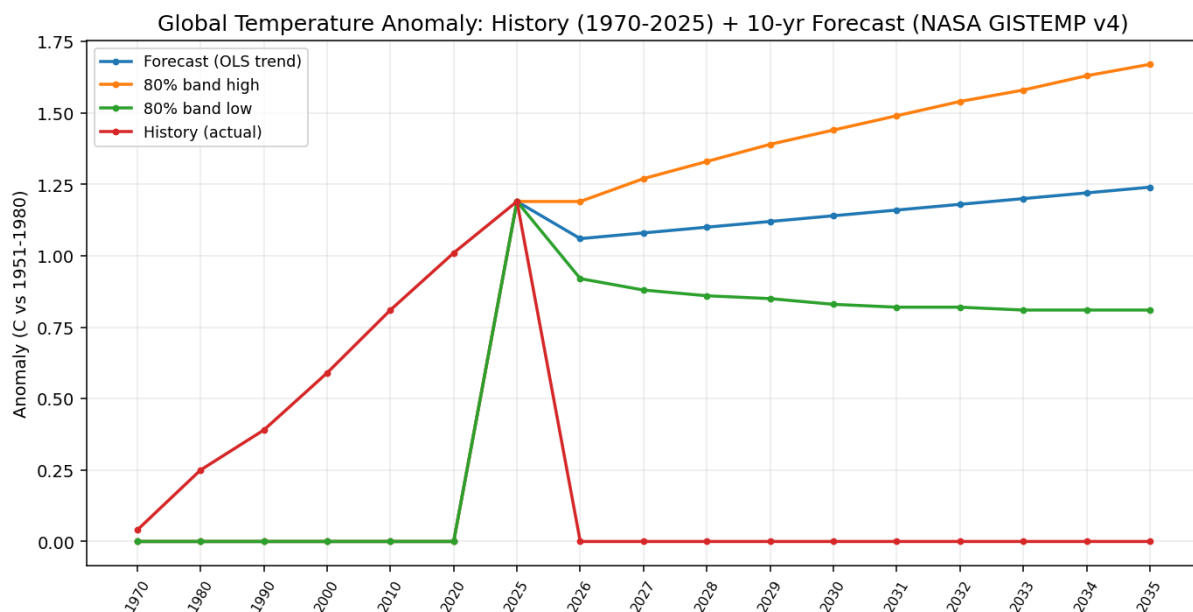
5. Notable events / inflections

The largest year-over-year moves in the record are:

Year	Annual (degC)	YoY change	Verified cause
2023	+1.17	+0.28	Record warmth: strong El Niño + falling sulfate aerosols
2024	+1.29	+0.12	Warmest year on record, ~1.28-1.29 degC
1977	+0.18	+0.28	Late-1970s warming step; boosted by volcanic aerosols

NASA's 2023 explainer notes the Hunga Tonga eruption raised stratospheric water vapor (warming) partly offset by volcanic sulfate aerosols (cooling) [Five Factors to Explain the Record Heat in 2023 - <https://science.nasa.gov/earth/earth-observatory/five-factors-to-explain-the-record-heat-in-2023-152313/>].

6. 10-year forecast



Forecast fit on the 1970-2025 sub-series (the modern, monotonic era) via OLS trend + 80% bands:

Year	Forecast (degC)	80% low	80% high
2026	1.06	0.92	1.19
2030	1.14	0.83	1.44
2035	1.24	0.81	1.67

Model vs naive - honest assessment: the seasonal-naive baseline (repeat last value = 1.19 degC) is competitive: the trend model's 80% band contains the naive value throughout, and the naive point estimate sits at the top of the band. Because annual anomalies are noisy and volcano/ENSO shocks are unmodeled, the OLS trend is not clearly better than naive - it should be read as a central-tendency path, not a precise prediction. (A full-history fit gives MAPE ~102%, unreliable because early anomalies cross zero; this is why the modern 1970+ window was used.)

7. Limitations

- Baseline ambiguity: the 1951-1980 baseline is inferred from the data's centering and NASA convention; the table carries no metadata row confirming it.
- Unmodeled drivers: ENSO (El Niño/La Niña), volcanic eruptions, and solar cycles cause +/-0.1-0.3 degC swings not captured by a linear trend - a key reason the forecast only ties the naive baseline.
- Future row present: 2026 is blank; forecasts beyond 2025 carry widening uncertainty (band grows from +/-0.14 to +/-0.43 degC by 2035).
- Single-vintage file: no uncertainty/error columns are included, so NASA's published per-year uncertainty margins could not be integrated.

8. What to do

- Treat this as authoritative climate data, not a client-business series - it is NASA GISTEMP v4 in degC on a 1951-1980 baseline. Cite it as such and don't rescale it without converting the baseline.
- Headline for stakeholders: +1.28 degC of warming between the 1880-1909 and 2016-2025 windows; 2024 was the hottest year at +1.29 degC.
- Use the forecast directionally: expect the trend to keep climbing toward ~+1.2 degC by 2035, but present it with the 80% band and note the model does not beat naive - the honest message is "continued warming within a wide range," not a point call.
- Next step if precision matters: add NASA's published uncertainty columns and an ENSO index to separate the human-caused trend from natural variability.